Supplementary Material S4
Complete Stratified Synthesis of the Corpus

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements Table 22 in the manuscript (studies of greatest translational traceability, Section 6.3). Section A presents the 30 eligible studies organised by clinical domain (D1–D4); Section B presents the 15 contextual studies outside the eligible set; Section C presents the systematic longitudinal tracking audit and the multi-label coding matrix for all 30 eligible studies.

The internal # column corresponds to the corpus order shared with Supplementary Material S2 (C prefix for contextual evidence), not to bibliographic reference numbers. Gaps in the sequence are not numbering errors: numbers were assigned before the final integrity and eligibility verification and are not reused. Ahmad et al. (2024), retracted after corpus finalisation (17 August 2026, *Healthc Technol Lett* 2026;13(1):e70095), is excluded with retroactive effect. The 8 retracted or withdrawn records do not appear in this document.

Version V3.6 reconstructed the screening (Supplementary Material S5). Seventeen studies were added (identifiers 29 to 51); their coding is first-reviewer coding verified by the second reviewer, and ND marks fields that could not be confirmed in the full text. Reed et al. (2021) and Almeida et al. (2019) left the eligible corpus because they do not use or evaluate an automated analysis of the face or gaze (criterion CI(1) made explicit in V3.6). The audit of criterion CI(3) against the SCImago edition of each publication year (`data/sjr_verification.csv`) moved Rudovic et al. (2018), Dapogny et al. (2019), Rudovic et al. (2019), Sholikah et al. (2024) and Sariyanidi et al. (2025) to contextual evidence (Section B), because their venues had no SJR quartile in that year. The full-text check of 12 November 2026 moved Cicceri et al. (2026) to contextual evidence under CE(3), because it analyses only hand keypoints to detect pointing gestures, and corrected the rows of Hassan et al. (2025), Silva et al. (2021) and Voss et al. (2019); the section of each article that supports every value is recorded in `data/fulltext_coding.csv`.

Section A. Eligible corpus (30 studies)

#	Study	Clinical task	Dataset	Population / Setting	Architecture	Metrics	Data prov.
Domain D1: Automated facial expression (10 studies)							
1	Johnson et al. (2025)	Emotion recogn.	AffectNet + clinical	N=48 (15 ASD) / Lab./Ecol.	EfficientNet-v2	ACC 67.8%, F1 67.8%[i]	Clinical
8	Radočaj & Mart. (2025)	Emotion recogn.	FERAC	ASD children / Lab.	Swin Transformer	ACC 80%, F1 78.9%	Partial prov.
9	Washington et al. (2022)	Emotion recogn.	CAFE Sub. A + clinical	Children / Lab./Ecol.	ResNet-152	Bal.ACC 79.1%, F1 78%	Clinical
21	Aljbori et al. (2025)	Emotion recogn.	Kaggle ASD	ASD children / Lab.	3D-CNN + LSTM	ACC 99.01%[g], F1 98.6%	Web
28	Leo et al. (2018)	Facial prod. ASD	CK+ + clinical (N=17)	ASD children / Clinical	HOG + SVM + CNN	TP 85% (CK+)[b]	Clinical
29	Kiejdo et al. (2025)	Emotion recogn.	EMBOA (99 recordings)	ASD 0–12 / Robot th.	BiLSTM + late fusion	ACC 68%, F1 68%	Clinical
30	Wang et al. (2025)	Emotion recogn.	Private (N=33)	ASD 4–5 / Rehab. classroom	CvT + MS-CAM fusion	ACC 90.73%	Clinical
31	Kurian & Tripathi (2024)	Emotion recogn.	SSBD (75 videos) + YouTube	Children (web videos) / Ecol.	m_AutNet + WGAN	ACC 88.25%	Web
32	Manelis-Baram et al. (2025)	FER agreement	NADI (N=100)	ASD + TD 2–7 / Clinical	iMotions, FaceReader, Py-Feat	r=0.37–0.75	Clinical
48	Kalantarian et al. (2019)	Emotion labelling	Game video (N=8 ASD)	ASD boys 8.5 y / Mobile game	Classifier ensemble	56–94%[n]	Clinical
Domain D2: Affect and therapeutic engagement (3 studies)							
4	Lecciso et al. (2021)	Therapeutic interv.	Private (N=24)	ASD children / Therapy	OpenFace	g=0.52	Clinical
34	Shi et al. (2022)	Arousal / valence	Private in-home (N=4)	ASD mean 4.6 / In-home robot	XGBoost + s-DA	AUROC 0.92 / 0.86	Clinical

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#	Study	Clinical task	Dataset	Population / Setting	Architecture	Metrics	Data prov.
35	Silva et al. (2021)	Real-time recogn.	Own non-ASD (N=43)	ASD N=6, 8–9 / Robot th.	SVM (RBF)	ACC 93.63%[o]	Non-clinical
Domain D3: Emotion recognition by the child (6 studies)							
2	He et al. (2019)	Eye-tracking	Private (N=42)	ASD preschoolers n=21 / Clinical	Eye-tracking	r=0.619	Clinical
39	Voss et al. (2019)	Wearable interv. (RCT)	Trial data (N=71)	ASD 6–12 / Home	Glass FER classifier	VABS-II +4.58	Clinical
40	Fan et al. (2025)	Emotion learning game	CAFE + FER-2013	ASD N=12, 4–7 / School	CNN	Feedback ACC 97.6%[p]	Public gen.
41	Wall et al. (2025)	App interv., ET + EEG	Private (N=35)	ASD 15 + NT 20, 6–12 / ND	Eye-tracking + ERP	Z=−3.98	Clinical
45	Carter Leno et al. (2020)	Attention to fear	Cohort ET (N=46)	ASD adolescents / ND	Eye-tracking	Assoc. with CU traits	Clinical
51	Bochet et al. (2021)	Emotional face expl.	Private ET (N=46)	ASD 24 + TD 22, 1.8–5.8 / ND	Eye-tracking	p=0.004	Clinical
Domain D4: Screening or diagnosis of ASD (11 studies)							
5	Shaik et al. (2025)	ASD detection	Kaggle ASD	Children / Lab.	Ensemble CNN	ACC 84%, F1 84.2%	Web
13	Hassan et al. (2025)	ASD level assess.	DREAM (N=61; 54 analysed)	ASD children 3–6 / Robot th.	Boosted RF + VGG16, majority vote	ACC 84.0%, F1 84.0% (test)[l]	Public cur.
19	Alam et al. (2025)	ASD detection	Kaggle + YTUIA; UIFID test	Children / Lab.	ASD-UANet	ACC 96%	Web
20	Hwang et al. (2024)	DD screening	Private (N=89; 33 DD, 56 TD)	34–77 months / Mobile	LSTM + 2D-FAN	ACC 88%, F1 80%	Clinical
23	Vedantham et al. (2025)	ASD detection	Kaggle	Children / Lab.	GWO-GEfficientNet	ACC 96.5%, F1 96.4%	Web
24	Hosney et al. (2024)	ASD detection	Kaggle (co-auth.)[d]	Children / Lab.	YOLOv8 + DCNN	ACC 97.2%, F1 93%	Web
33	Chen et al. (2024)	ASD identification	Private (N=81)	ASD 41 + TD 40, 4–6 / School	LSTM + SENet	ACC 88.74%	Clinical
36	Alvari et al. (2021)	Early smile markers	Home videos (N=33)	ASD 18 + TD 15, 6–12 m / Home	OpenFace AUs	MANOVA $p < 0.0001$	Clinical
37	Carpenter et al. (2020)	Toddler screening	Private (N=104)	Toddlers (22 ASD) / Tablet	CV + LASSO logistic	AUC 0.62–0.73	Clinical
38	Pliska et al. (2025)	Digital screening	Private (N=48)	ASD 24 + TD 24, 6–11 / Clinic, school	Decision trees	ACC 81.25%	Clinical
49	Han et al. (2022)	ASD identification	Private (N=90)	ASD 40 + TD 50, 3–6 / ND	MMSDAE fusion	ACC 95.56%, Sens. 92.5%[q]	Clinical

Section B. Contextual evidence outside the eligible set (15 studies)

Five infringe CE(3) (exclusively non-facial data), three infringe the age criterion (exclusively adult participants) and six infringe CI(3) (venue without SJR quartile in the year of publication).

#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data prov.
Adult participants (3 studies)								
C1	El Rhatassi et al. (2024)	Emotion recogn.	FER-2013 + Kaggle autism	Autistic adults	Lab.	CNN vs ViT	ACC 65–74%	Public gen.
C2	El Rhatassi et al. (2025)	Real-time recogn.	CK+ + HeyGen AI	Autistic adults	Lab.	CNN×3 + LSTM + CC	ACC val. 53%	Mixed
C3	Levy et al. (2024)	Wearable system	FER-2013	ASD adults (N=10)	Ecol.	CNN + HW wearable	Usability	Public gen.
Non-facial data only, CE(3) (6 studies)								
C4	Nagae & Lee (2022)	Therapy + EDA	Private EDA (N=11)	DD children	Robot th.	EDA + wavelet + SVM	Conf. matrix	Clinical

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#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data prov.
C5	Papaioannou et al. (2021)	EEG cogn. load	EEG (N=63)	ASD + ADHD adults	Clinical	MSE + PLSC	ANOVA	Clinical
C6	Xu & Ju (2023)	VBM brain	ABIDE (N=203)	ASD 8–12 (n=98)	Clinical	VBM	GMV+ADOS corr.	Public cur.
C7	O'Sullivan et al. (2023)	Speaker diariz.	Clin. trial audio (N=72)	ASD 5–45	Home	Pyannote + xvector	F1 76.5%	Clinical
C8	Alsubai (2025)	ASD detect. (tab.)	Kaggle screening, text, clinical	12–36 m; adults 60–96	Screening	Transformer MDNCT	ACC 94.7%	Web
C15	Cicceri et al. (2026)	Pointing gestures	Private (N=6)	ASC 23–63 m	Clinical	YOLOv5 + hand DNN	mAP 88.6%	Clinical
Venue without SJR quartile in the year of publication, CI(3) (6 studies)								
C9	Akın-Bülbül et al. (2024)	ET screening	Private ET	Infants 18–36 m	Clinical	KNN + features	ACC 81.45%	Clinical
C10	Rudovic et al. (2018)	Affect / therapy	MIT private (N=35)	ASD children	Robot th.	PPA-net	ICC ∼60%	Clinical
C11	Dapogny et al. (2019)	Facial expr.	JEMImE (N=157)	Children 6–11	Clin. game	Rand. Forests + PCRF	ACC 81.9%	Public cur.
C12	Rudovic et al. (2019)	Active learning	MIT private (N=35)	ASD children	Robot th.	4 LSTMs + Deep RL	ACC 82.3%, F1 52.6%[k]	Clinical
C13	Sholikah et al. (2024)	Real-time recogn.	FER-2013	N=3 ASD (test)[a]	Mobile	VGG-16	ACC 91%, F1 91%	Public gen.
C14	Sariyanidi et al. (2025)	Autism pred.	CHOP clinical (N=42)	ASD + TD	Clinical	Dict. Learn. + SVM	Fig.[c]	Clinical

Section C. Longitudinal tracking audit and multi-label coding matrix

Section C.1 presents the systematic longitudinal tracking audit for all 30 eligible studies, evaluating the presence of intraindividual temporal series, progress outcomes, test-retest reliability, and responsiveness to change. Section C.2 presents the multi-label coding matrix, documenting the objective variable type, output class structure, and training representation for each study.

The principal finding confirmed by the audit is that no study in the eligible corpus meets the operational definition of session-to-session longitudinal tracking. The closest, Silva et al. (2021), records in an automated way the correct answers and response time of each child across seven robot sessions and presents the evolution of five children descriptively, without quantifying change, test-retest reliability or responsiveness; the previous coding of one session was corrected against the full text. The other studies with multiple recorded sessions do not fill this gap: Hassan et al. (2025) (61 children and 3,121 sessions in DREAM, 54 children analysed) uses the session data to predict the ASD level and the ADOS score of each child and does not quantify intraindividual change trajectories, test-retest reliability, or responsiveness. Shi et al. (2022) (4 children, 19 in-home sessions in total) predict arousal and valence within each session with a chronological split by child, but do not quantify change across sessions. Lecciso et al. (2021), Voss et al. (2019), Fan et al. (2025) and Wall et al. (2025) use pre-post designs; they compare measurements before and after an intervention at group level, with an additional 6-week follow-up in Voss et al. (2019), but do not report intraindividual session-by-session series. Longitudinal therapeutic monitoring therefore remains an open gap.

C.1. Longitudinal tracking audit (30 eligible studies)

Columns: *Sessions*: number of distinct measurement points or sessions in the study design; *Subject unit*: unit of analysis (individual vs group); *Intraindiv. series*: intraindividual temporal series across sessions (Yes / No); *Progress outcomes*: session-to-session progress indicators (Yes / No); *Test-retest*: test-retest reliability coefficient reported (Yes / No); *Responsiveness*: sensitivity to change across time or intervention (Yes / No).

#	Study	Sessions	Subject unit	Intraindiv. series	Progress outcomes	Test-retest	Responsiveness
Domain D1 (10 studies)							
1	Johnson et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
8	Radočaj & Mart. (2025)	1 (cross-sect.)	Group	No	No	No	No
9	Washington et al. (2022)	1 (cross-sect.)	Group	No	No	No	No
21	Aljbori et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
28	Leo et al. (2018)	1 (cross-sect.)	Group (N=17)	No	No	No	No
29	Kiejdo et al. (2025)	99 recordings	Group	No	No	No	No
30	Wang et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
31	Kurian & Tripathi (2024)	1 (web videos)	Group	No	No	No	No
32	Manelis-Baram et al. (2025)	1 (ADOS session)	Group (N=100)	No	No	No	No
48	Kalantarian et al. (2019)	1 (game session)	Group (N=8)	No	No	No	No
Domain D2 (3 studies)							
4	Lecciso et al. (2021)	2 (pre-post)	Individual	Pre-post only[e]	No	ND	ND
34	Shi et al. (2022)	19 in-home sessions[r]	Individual (N=4)	No[r]	No	No	No
35	Silva et al. (2021)	7 robot sessions	Individual (N=6; 5 interpretable)	Yes (descriptive)[u]	Yes (descriptive)[u]	No	No
Domain D3 (6 studies)							
2	He et al. (2019)	1 (cross-sect.)	Individual	No	No	No	No
39	Voss et al. (2019)	3 (intake, 6 wk, 12 wk; RCT)	Group (N=71)	Pre-post with follow-up[e]	No	ND	ND
40	Fan et al. (2025)	3–7 + pre/post (3 wk)	Group (N=12)	Pre-post only[e]	No	ND	ND
41	Wall et al. (2025)	2 (pre-post, 2 weeks)	Group (N=35)	Pre-post only[e]	No	ND	ND
45	Carter Leno et al. (2020)	1 (cross-sect.)	Group (N=46)	No	No	No	No
51	Bochet et al. (2021)	1 (cross-sect.)	Group (N=46)	No	No	No	No
Domain D4 (11 studies)							
5	Shaik et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
13	Hassan et al. (2025)	3,121 sessions total	Group (N=54 of 61)	No[f]	ND	ND	ND
19	Alam et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
20	Hwang et al. (2024)	1 (screening)	Group (N=89)	No	No	No	No
23	Vedantham et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
24	Hosney et al. (2024)	1 (cross-sect.)	Group	No	No	No	No
33	Chen et al. (2024)	1 (cross-sect.)	Group (N=81)	No	No	No	No
36	Alvari et al. (2021)	1 (home videos)	Group (N=33)	No	No	No	No
37	Carpenter et al. (2020)	1 (screening)	Group (N=104)	No	No	No	No
38	Pliska et al. (2025)	1 (screening)	Group (N=48)	No	No	No	No
49	Han et al. (2022)	1 (cross-sect.)	Group (N=90)	No	No	No	No

C.2. Multi-label coding matrix (30 eligible studies)

Columns: *Obj. variable*: type of target variable; *Output classes*: number of output categories; *Training repr.*: primary training representation modality; *Stratified val.*: stratified or subject-independent validation reported; *Multi-label*: whether the study reports multi-label (simultaneous assignment of $\geq$2 non-exclusive labels) as opposed

to multi-class or binary single-label classification.

#	Study	Obj. variable	Output classes	Training repr.	Stratified val.	Multi-label
Domain D1 (10 studies)						
1	Johnson et al. (2025)	Multi-class emotion	7 (basic emotions)	CNN / AffectNet+clinical	No	No
8	Radočaj & Mart. (2025)	Multi-class emotion	4 (natural, fear, joy, anger)	Transformer / FERAC	Yes[j]	No
9	Washington et al. (2022)	Multi-class emotion	7 (basic emotions)	CNN / CAFE subset	Yes	No
21	Aljbori et al. (2025)	Multi-class emotion	6 (basic emotions)	3D-CNN + LSTM / Kaggle	No	No
28	Leo et al. (2018)	Multi-class emotion	7 + ASD prod.	HOG+CNN / CK+ + clinical	N/A	No
29	Kiejdo et al. (2025)	Multi-class emotion	6 (basic emotions)	BiLSTM / FaceReader + biosignals	Yes (subject-disjoint)	No
30	Wang et al. (2025)	Multi-class emotion	4	CvT / face + speech	Yes (subject-wise)	No
31	Kurian & Tripathi (2024)	Multi-class emotion	4 (happy, sad, neutral, meltdown)	CNN + WGAN / SSBD + YouTube	No[s]	No
32	Manelis-Baram et al. (2025)	Expression probabilities	N/A	Off-the-shelf FER / ADOS video	N/A	No
48	Kalantarian et al. (2019)	Multi-class emotion	7 (6 emotions + neutral)	Commercial classifier ensemble / game frames	N/A	No
Domain D2 (3 studies)						
4	Lecciso et al. (2021)	Continuous affect	N/A (AU intensity)	OpenFace / clinical	N/A	No
34	Shi et al. (2022)	Binary arousal; binary valence	2 + 2	XGBoost / multimodal features	Yes (session-wise)	No
35	Silva et al. (2021)	Multi-class emotion	6	SVM / 3D landmarks + AUs	ND[o]	No
Domain D3 (6 studies)						
2	He et al. (2019)	Correlation (continuous)	N/A	Eye-tracking	N/A	No
39	Voss et al. (2019)	Multi-class emotion	8	Embedded classifier / ND	N/A	No
40	Fan et al. (2025)	Multi-class emotion	7 (FER-2013); 4 evaluated	CNN / CAFE + FER-2013	No (image-level split)	No
41	Wall et al. (2025)	Group comparison (continuous)	N/A	Eye-tracking + ERP	N/A	No
45	Carter Leno et al. (2020)	Association (continuous)	N/A	Eye-tracking	N/A	No
51	Bochet et al. (2021)	Group comparison (continuous)	N/A	Eye-tracking	N/A	No
Domain D4 (11 studies)						
5	Shaik et al. (2025)	Binary (ASD / TD)	2	CNN ensemble / Kaggle	Yes[m]	No
13	Hassan et al. (2025)	ASD level[f]	2 (ASD, autism)	Boosted RF + VGG16 / DREAM	Yes[f]	No
19	Alam et al. (2025)	Binary (ASD / TD)	2	CNN ensemble / Kaggle+YTUIA	Yes	No
20	Hwang et al. (2024)	Binary (DD / TD)	2	LSTM+2D-FAN / clinical	Yes	No
23	Vedantham et al. (2025)	Binary (ASD / TD)[h]	2	CNN hybrid / Kaggle	No	No
24	Hosney et al. (2024)	Binary (ASD / TD)	2	YOLO+CNN / Kaggle	No	No
33	Chen et al. (2024)	Binary (ASD / TD)	2	LSTM + SENet / facial descriptors	Yes (by participant)	No
36	Alvari et al. (2021)	Group comparison (continuous)	N/A	OpenFace AUs	N/A	No
37	Carpenter et al. (2020)	Binary (ASD / non-ASD)	2	LASSO logistic / CV features	Yes (leave-one-out)	No
38	Pliska et al. (2025)	Binary (ASD / TD)	2	Decision trees / gaze + task	No[t]	No
49	Han et al. (2022)	Binary (ASD / TD)	2	MMSDAE / EEG + eye-tracking	Yes (10-fold, subject-indep.)	No

Notes.

a. 3 ASD participants in Sholikah et al. (2024) involved in a usability test, not in metric computation.

b. TP 85% for Leo et al. (2018) corresponds to the CK+ public dataset of typically developing adults, not to the 17 ASD children.

c. Fig.: metrics reported in graphical format only, without numerically extractable values. Applies to Sariyanidi et al. (2025).

d. (co-auth.): Kaggle dataset created by a co-author of Hosney et al. (2024), constituting a documented conflict of interest.

e. Lecciso et al. (2021) employs a pre-post design comparing two time points (g=0.52) but does not report intraindividual session-by-session series, test-retest reliability, or responsiveness. The same applies to the pre-post designs of Voss et al. (2019), a randomised controlled trial with 51% adherence, Fan et al. (2025),

whose 12 children had three to seven 30-minute sessions over three weeks and whose system logs compare the star rate of the first and last training days only (sections 4.2 and 4.4 of the article), and Wall et al. (2025).

f. Hassan et al. (2025) use the 3,121 robot-enhanced therapy sessions of the 61 children in the DREAM dataset to predict the ASD level (classification) and the ADOS score (regression, RMSE 1.78, R^2 0.74). Because the outcome is the severity of the condition and not engagement or affect during the session, the study is assigned to domain D4. After cleaning, 54 of the 61 children are analysed; 46 participants train the models and 8 are held out as unseen test data, and a stratified cross-validation generates the training scores (Result and Analysis section). Because DREAM contains no participant without an ASD diagnosis, the ADOS cut-offs yield two classified levels: ASD (Asperger syndrome and PDD) and autism (Data Pre-Processing and Segmentation section). The study does not quantify intraindividual change trajectories.

g. Aljbori et al. (2025) report 99.01% accuracy in their results table and 99.82% in the abstract; the table value is retained here.

h. Vedantham et al. (2025) classify autistic versus non-autistic facial images; an auxiliary facial expression dataset is relabelled into three classes (positive, negative, neutral) without a stated mapping to the autism labels.

i. The 67.8% reported by Johnson et al. (2025) corresponds to seven-class classification on the 4,000 AffectNet validation images (precision 67.9%, recall 67.8%; table 3 of the article), not to the 48 participants of the clinical study (15 with ASD, school and university students), who recognised expressions in original and enhanced images.

j. Radočaj and Martinović (2025) state an 80/20 stratified train-validation split, while the image counts they report are inconsistent (691/79 in one section and 615/155 in another).

k. Best values after adaptation in Rudovic et al. (2019): multimodal Q-learning with model-level fusion, three engagement levels (Tables 2 and 3 of the article), with training and test children kept separate.

l. Hassan et al. (2025): the 84.0% accuracy, 85% precision, 86% recall, 84% F1-score and micro-averaged AUC of 0.93 correspond to the boosted 3D biomarker model on the 8 held-out participants (Table III of the article); the 95.6% accuracy and 92.8% F1-score previously coded are the training cross-validation scores (Table II); the majority vote with the gaze saliency maps reports 97.36% accuracy and an F1-score of 95.56% without stating the set on which it is computed (Table VI).

m. Validation scheme verified in the full text: Shaik et al. (2025) use stratified 5-fold cross-validation together with a stratified 80/20 split.

n. Kalantarian et al. (2019) label the frames of game videos of 8 boys with ASD (one session each) with commercial emotion classifiers combined in an ensemble, over 7 categories (6 emotions and neutral); the share of correctly labelled frames is 94% for disgust, 81% for neutral, 92% for surprise and 56% for scared, without significant improvement for angry and happy.

o. Silva et al. (2021) train on 43 participants without ASD (11 adults and 32 typically developing children) and report 93.63% with 10-fold cross-validation offline, without stating the partition unit, and 88.3% in real time with adults; the six children with ASD took part in a pilot test.

p. The 97.6% of Fan et al. (2025) is the accuracy of the real-time feedback of the system judged by specialists on 330 clips; the classifier was trained on data from children without ASD.

q. Han et al. (2022) report 86.67% accuracy for eye-tracking and 81.11% for EEG; the fused model reaches 95.56% accuracy, 92.5% sensitivity and 98% specificity under 10-fold subject-independent stratified cross-validation.

r. Shi et al. (2022) record 19 in-home sessions in total with a socially assistive robot and predict arousal and valence within each session with a chronological split by child; change across sessions is not quantified.

s. Kurian and Tripathi (2024) report a test accuracy of 88.25% on a hold-out set of 80 samples; separation of children between training and test is not reported.

t. Pliska et al. (2025) report no cross-validation or hold-out set; the accuracy is computed on the study sample, and the authors warn of overfitting.

u. Silva et al. (2021) apply the system with six children with ASD during seven sessions with the ZECA robot; the robot's facial expression recognition determines whether each imitation or production is correct, and the article plots the correct and incorrect answers and tabulates the response time of each child per session (Figures 6 to 10 and Tables 7 to 10). The series is descriptive: change is not quantified, and test-retest reliability and responsiveness are not evaluated.

1. The internal # column uses the corpus order shared with Supplementary Material S2 for eligible studies and the C prefix for contextual evidence. It does not correspond to bibliographic reference numbers, and gaps in the sequence are not numbering errors.
2. Retracted or withdrawn records do not appear, among them Güngör et al. (2025), Mahmood et al. (2025), Kumar et al. (2024) and Ahmad et al. (2024). Ahmad et al. (2024) was retracted on 17 August 2026 (*Healthc Technol Lett* 2026;13(1):e70095). All eight retracted or withdrawn records are listed in Supplementary Material S5 and documented in Section 2.7 of the manuscript.
3. Bal.ACC for Washington et al. (2022): balanced accuracy 79.1% on CAFE subset with human agreement ≥60%; full CAFE set value is 66.9%.
4. Data prov.: Non-clinical marks a training set built by the authors with participants without ASD.
5. Contextual studies in Section B were appraised with MMAT in Supplementary Material S2 (Section B) but do not participate in the aggregate MMAT distribution of the manuscript.